

0.5-V Analog Circuit Techniques and Their Application in OTA and Filter Design

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Abstract—We present design techniques that make possible the operation of analog circuits with very low supply voltages, down to 0.5 V. We use operational transconductance amplifier (OTA) and filter design as a vehicle to introduce these techniques. Two OTAs, one with body inputs and the other with gate inputs, are designed. Biasing strategies to maintain common-mode voltages and attain maximum signal swing over process, voltage, and temperature are proposed. Prototype chips were fabricated in a 0.18- μm CMOS process using standard 0.5-V V_T devices. The body-input OTA has a measured 52-dB DC gain, a 2.5-MHz gain-bandwidth, and consumes 110 μW . The gate-input OTA has a measured 62-dB DC gain (with automatic gain-enhancement), a 10-MHz gain-bandwidth, and consumes 75 μW . Design techniques for active-RC filters are also presented. Weak-inversion MOS varactors are proposed and modeled. These are used along with 0.5-V gate-input OTAs to design a fully integrated, 135-kHz fifth-order elliptic low-pass filter. The prototype chip in a 0.18- μm CMOS process with V_T of 0.5-V also includes an on-chip phase-locked loop for tuning. The 1-mm² chip has a measured dynamic range of 57 dB and draws 2.2 mA from the 0.5-V supply.

Index Terms—Active filters, analog integrated circuits, body bias, low voltage, operational amplifiers, varactors.

I. INTRODUCTION AND MOTIVATION

CONTINUING technology feature-size scaling requires a proportional downscaling of the supply voltage to maintain device reliability. At the same time, a relatively large threshold voltage (V_T) needs to be maintained to limit the OFF current [1] in transistors. These two factors pose significant challenges for the design of analog circuits in future standard digital CMOS processes. Charge pump techniques are sometimes used to boost on-chip voltages beyond the supply; this may lead to reliability problems in some cases. However, a number of techniques can make possible true low-voltage operation without voltage boosting; these include the use of weakly inverted devices where feasible, the use of the body node as a fourth control or signaling terminal, and the use of circuit topologies avoiding stacked transistors. To investigate the synergistic use of such techniques, we have designed two operational transconductance amplifiers (OTAs) and an automatically tuned fifth-order low-pass filter, and have pushed their operation to supply voltages as low as the V_T of the standard devices, in our case, 0.5 V [2], [3].

In Section II, we discuss two different true low-voltage OTA design techniques using standard CMOS technology, first using

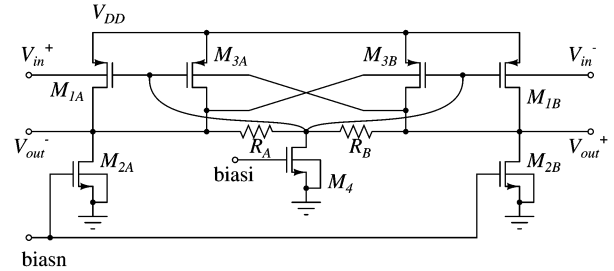


Fig. 1. Fully differential body-input gain stage with local common-mode feedback.

the body nodes of the MOS devices to apply the signal with biasing accomplished through the gates, and then using the gate nodes of the MOS device for the signal with biasing through the body. We contrast the two techniques using prototypes on a CMOS 0.18- μm process with measurement results. In Section III, we propose and analyze a low-frequency weak-inversion 0.5-V MOS varactor for use in active varactor-R filters. In Section IV, we demonstrate the design of a filter using gate-input OTAs and the varactors as building blocks. The prototype chip has been extensively characterized and experimental results for different chips, different supply voltages, and different temperatures are presented. Note that future CMOS technologies will offer much faster transistors that also have a lower V_T compared to 0.18- μm transistors. The techniques demonstrated here are thus expected to yield higher performance once these technologies become available [1].

II. 0.5-V FULLY DIFFERENTIAL OPERATIONAL TRANSCONDUCTANCE AMPLIFIERS

For any high-performance application that requires high bandwidth or sampling rate, MOS devices are biased in strong inversion, i.e., $(V_{GS} - V_T) > 0.2$ V. The devices act as transconductors or current sources as long as $V_{DS} \geq V_{DS,sat}$. A good estimate of $V_{DS,sat}$ at the edge of strong inversion or in weak and moderate inversion is about 0.15 V [4]. So, in any region of operation, we need to maintain V_{DS} of at least 0.15 V, irrespective of the device V_T . The common-source configuration thus has the potential to operate at supply voltages of 0.5 V.

Forward biasing of the body-source junction has been applied in low-voltage digital circuits [5]–[8] and it is applied here to lower the V_T of the transistors. We typically apply a forward bias of about 250 mV, which results in a lowering of the V_T by about 50 mV. In the context of 0.5-V operation, the risk of forward-biasing the junctions is minimized since parasitic bipolar devices cannot be activated even when the full power supply is

Manuscript received April 15, 2005; revised July 15, 2005. This work was supported in part by Analog Devices.

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Digital Object Identifier 10.1109/JSSC.2005.856280